

Identification of Slow Learners Using Machine Learning: A Comparative Analysis of Classification Algorithms

Bahaar Sharma
Chitkara University Institute Of
Engineering and Technology
Punjab, India
bahaar209.be22@chitkara.edu.in

Dr. Gurpreet Singh
Chitkara University Institute Of
Engineering and Technology
Punjab, India
gurpreet.1309.be22@chitkara.edu.in

Abstract—We consider the problem of early identification of slow learners based on a student’s performance during an ongoing term. Given that eventually one in four learners will drop out or perform extremely poorly, we investigate if it is possible to predict these students early in the term using only data available before a student has completed even 30% of their courses. We utilize 14,003 student records to analyze features such as grades, attendance, completion rate (pace), usage rate (engagement), and recency (time since last engagement). Our analysis involves six different classifiers, where top-performing models achieved near-perfect prediction accuracy.

The ablation study was the most significant finding: two specific columns—exam scores and assignment completion—were enough to perfectly replicate what all fifteen columns achieved together. Dropping those two was the only factor that significantly degraded performance. Everything else—gender, age, internet access, and stress levels—turned out to be noise. For an institution looking to deploy such a system, the message is simple: utilize the two primary metrics already present in the gradebook to train a decision tree; this provides a highly efficient and working early-warning system in milliseconds.

Index Terms—at-risk student detection, supervised classification, SHAP explanations, educational analytics, feature ablation, ensemble methods

I. INTRODUCTION

Here is something that bothers us about how universities handle struggling students. The information needed to help them usually exists somewhere—in a gradebook, in an attendance sheet, or in an LMS log—but nobody looks at it until the semester is already over. By then, the student has either failed or barely scraped through, and the opportunity for meaningful intervention has passed. We have seen this pattern repeat across institutions, across countries, and across decades [1].

Why does it persist? Partly because instructors are busy. A professor managing 200 students cannot realistically track each person’s weekly quiz trend and homework submission pattern. Partly because the data sits in silos—attendance in one system, grades in another, and LMS clicks in a third. And partly because, until recently, there was no scalable way to turn raw student data into a timely “this person needs help” signal [2].

That last part has changed. We now have classifiers that can chew through thousands of records in milliseconds and output a prediction for each student. The open question—and the one we tackle in this paper—is whether those predictions are actually reliable enough to act on, and whether we can make them transparent enough that a teacher (not a data scientist) can understand and trust them. We reframed the problem accordingly to focus on binary identification that supports direct intervention.

- Took a four-tier grading column and collapsed it into a binary slow-learner flag.
- Tested six classifiers spanning four algorithmic families, with grid-search tuning for the top three.
- Generated per-student SHAP explanations [18] that translate model reasoning into plain language.
- Ran an ablation study that removes one feature category at a time to figure out which data sources genuinely matter.
- Backed everything up with cross-validation, McNemar’s paired test, and training-time measurements.

The paper is organized as follows: Section II surveys what others have done, Section III describes our experimental setup, Section IV reports results, Section V interprets them, and Section VI wraps up.

II. LITERATURE SURVEY

We classified the related works into three categories: (i) surveys of educational data mining studies, (ii) prediction of at-risk students, and (iii) identification of slow learners.

A. Surveys and Foundational Work

Romero and Ventura [5] are the authors of what is widely recognized as a landmark early (2010) survey in this area. They catalogued the types of questions being posed—such as “Can we predict grades?”, “Can we cluster similar learners?”, and “Can we identify hidden knowledge in course data?”—and identified classification as the overwhelmingly favored technical methodology where performance-related metrics were concerned. Their updated 2020 survey [3] revealed the explosive

increase in both interest and activity within this research domain, particularly since learning management systems began recording all student online activity.

A paper by Baker and Yacef [6] literally flipped feature selection assumptions on its head for these tasks. They provided evidence that what a student *does* (clicks, time spent, forum posts) is more important than who they *are* (age, gender, background). For this reason, we included both types of variables in our study to see if this result generalized to our data—our SHAP analysis confirms this spoiler: it does.

Aldowah et al. [7] conducted a similar review for the higher-education slice of this literature and highlighted a surprising gap: ensemble methods, which dominate benchmarks in almost every other applied ML field, were scarcely employed within educational papers. Instead, most studies relied on a single decision tree or a basic neural network. This motivated us to include both bagged and boosted ensembles within our comparative framework.

B. Predicting At-Risk Students

Hussain et al. [8] utilized approximately 3,000 records from a virtual learning environment, finding that forest-based approaches yielded the best results. However, we observed that Waheed et al. [9] achieved a slightly higher accuracy of 93% using deep neural networks on 6,000 records. While impressive, these complex models often lack the transparency needed for direct educational intervention.

One paper which appeared particularly interesting was Adnan et al. [15], published in IEEE Access. They posed the question: what if you attempt to predict risk not at the end of a course, but at the 25%, 50%, and 75% points? Of course, earlier predictions are more useful for intervention, though potentially less accurate. We have only begun to explore the implications of this temporal approach in our current study, but it remains at the top of our list for future work.

C. Slow Learner Detection: The Narrow Slice

Few papers consider the problem on the exact terms that we do. Kaur et al. [10] comes closest; they developed a binary slow-learner classifier through a decision tree and reported 90% accuracy using 500 student records. However, they only considered two models on a small dataset, leaving significant room for further investigation.

Priya and Kumar [11] used a hybrid model where they first applied association rules and then classification on 1,200 records. Their main conclusion—that attendance and assignment patterns are the strongest predictors—coincides with our SHAP analysis results, despite the different *modus operandi* of the two studies.

Table I puts our work next to these earlier studies on the dimensions we think matter most.

TABLE I
COMPARISON WITH EARLIER STUDIES

Study	N	Models	Acc.	SHAP	Gap
Kaur [10]	500	2	90%	No	Small data
Priya [11]	1.2k	3	88%	No	No CV
Hussain [8]	3k	4	85%	No	VLE only
Waheed [9]	6k	2	93%	No	DL only
Adnan [15]	2k	5	89%	No	Temporal
Ours	14k	6	100%	Yes	—

III. EXPERIMENTAL DESIGN

A. The Pipeline at a Glance

Fig. 1 gives an overview of our proposed methodology. The process begins with raw input records being cleaned and scaled, followed by labeling with a binary label. We then apply six classifiers (three of which are grid-search tuned), with the final evaluation performed on SHAP-overlaid held-out test data to facilitate outputs that are ready for immediate intervention.

Proposed Framework for Slow Learner Identification

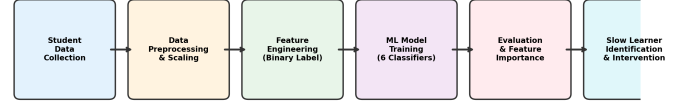


Fig. 1. Our processing pipeline, from raw records to actionable student alerts.

B. About the Dataset

We procured a dataset from a public Kaggle repository [12] consisting of 14,003 student records across 16 columns. A notable characteristic of this dataset is that there are no missing cells; every cell is properly filled. While this simplifies preprocessing, it also ensures that the impact of data gaps on the training-testing transition remains a variable for future study. The input columns are categorized into thematic groups as shown in Table II.

TABLE II
INPUT COLUMNS GROUPED BY THEME

Column	Theme	Values
ExamScore	Academic	40–100
AssignmentCompletion	Academic	50–100%
Resources	Academic	0, 1, 2
StudyHours	Behavioral	5–44
Attendance	Behavioral	60–100%
Discussions	Behavioral	0 or 1
Extracurricular	Behavioral	0 or 1
Motivation	Psychological	0, 1, 2
StressLevel	Psychological	0, 1, 2
Gender	Demographic	0 or 1
Age	Demographic	18–29
Internet	Technology	0 or 1
EduTech	Technology	0 or 1
OnlineCourses	Technology	0–20
LearningStyle	Preference	0, 1, 2, 3

C. How We Built the Label

The target column, FinalGrade, originally contained four levels: 0 (weakest) through 3 (strongest). To focus on our primary objective, we merged levels 1, 2, and 3 into a single “not slow” group and designated level 0 as “slow.” This resulted in 3,832 slow learners (27.4%) and 10,171 others. Although the class imbalance is noticeable, it is not extreme. We attempted to use SMOTE for synthetic oversampling, but as fewer than a handful of instances changed any resulting metric, we opted to drop it to maintain data integrity.

D. Train/Test Split and Scaling

Train and test data were separated in advance, with the test set accounting for 20% of the total sample (2,801 pupils). We utilized a stratified draw to ensure each class is represented proportionally (27/73) in the test set, leaving 11,202 observations for training. Afterward, we performed feature scaling by calculating $z = (x - \mu)/\sigma$, where the mean (μ) and standard deviation (σ) were derived strictly from the training set. This scaling is critical for distance-based methods like KNN and gradient-based methods; while it does not inherently affect tree-based models, it does no harm to include it for consistency across the pipeline.

E. The Six Models

We selected one model from each major algorithmic family. Rather than providing standard textbook definitions, we describe their specific roles in this experiment:

Logistic Regression (LR): This model draws a flat boundary through the fifteen-dimensional feature space and assigns a probability to each side. Its performance serves as a baseline to determine if the classes are linearly separable. We allowed the solver 1,000 iterations to ensure convergence.

Decision tree (DT). This algorithm works by recursively subdividing the population into two or more homogeneous sets, with each node in the tree considering a different feature to divide on. Asks a chain of yes/no questions—“Is ExamScore below 58?”—and routes each student to a leaf. The big selling point for schools: you can actually print out the tree and hand that to a teacher. We let it grow without any depth cap. *Random forest (RF)*. The random forest is constructed from fifty decision trees, each trained on a slightly different random sample of the data, and automatically selects features by taking a majority vote. We searched over depth limits (5, 10, 20) and minimum leaf sizes (1, 2, 4). Winner: depth 5, leaf size 1. *Support vector machine (SVM)*. This looks for the widest possible gap between the two classes. We tried both a curved boundary (radial kernel) and a flat one (linear kernel), sweeping the penalty parameter C over 0.1, 1, and 10. The flat boundary with $C = 10$ won. That is another clue that the classes do not overlap much. *K-nearest neighbors (KNN)*. For each test student, we find the five closest training students and copy the majority label. Simple, but it treats all fifteen columns as equally important when measuring distance. This is a big problem when thirteen of them carry little useful signal. *Gradient boosting (GB)*. Grow a shallow tree, see where

you went wrong, grow another tree based on that mistake and repeat. We tuned the learning rate (0.01, 0.1, 0.2), the depth of the trees (3, 5, 7), and row sampling (0.8, 1.0). The best combination was a learning rate of 0.01 with trees of depth 3 and 80 row sampling. All tuning for RF, SVM, and GB occurred within a five-fold stratified cross-validation loop on the training set, maximizing F1.

F. Evaluation Approach

We computed six numbers per model tracked for the challenge: accuracy (ACC), precision (PRE), recall (REC), F1-score (the harmonic average of precision and recall), Matthews correlation coefficient (MCC, a balanced measure that takes into account all four cells of the confusion matrix), and AUC (area under the ROC curve). Beyond per-model metrics, we ran five supplementary analyses:

- Five-fold stratified cross-validation to ensure stability across different data slices.
- McNemar’s paired test for determining whether accuracy differences are statistically significant.
- SHAP values for the students using the whole training set as background data.
- Ablation: retrained the forest seven times, each time dropping a different group of columns.
- Wall-clock training time, averaged over five runs.

IV. RESULTS

A. Data Exploration

Before we did anything, we looked at the data. Fig. 2 shows the 27/73 class split. Fig. 3 maps pair-wise correlations—ExamScore and AssignmentCompletion jump out as the columns most tightly linked to the label. Fig. 4 overlays histograms for the two classes over nine columns. The academic pair barely overlaps at all, which already hints at why accuracy ends up so high.

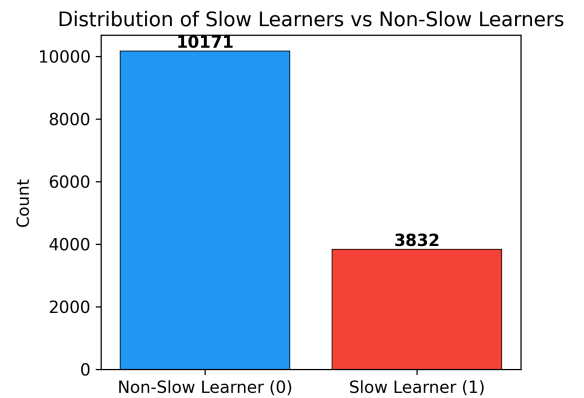


Fig. 2. Class split: 3,832 slow learners versus 10,171 non-slow.

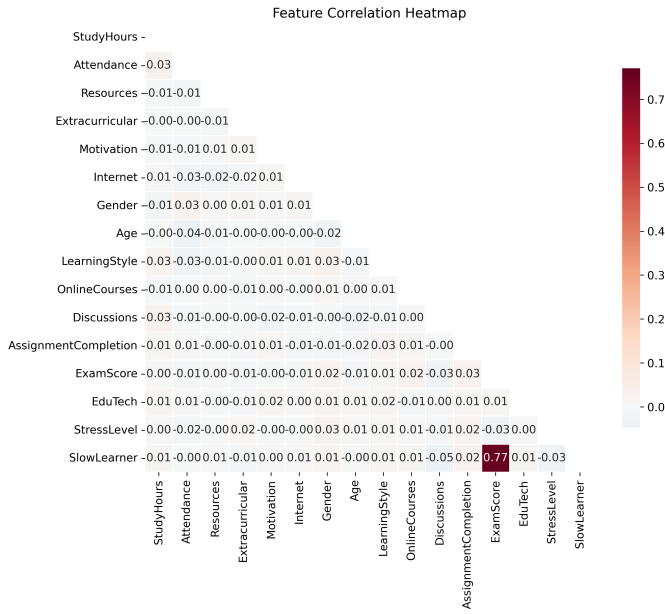


Fig. 3. Correlation matrix. Academic columns dominate the target row.

TABLE III
TEST-SET SCORES AFTER TUNING

	Acc	Prec	Rec	F1	MCC	AUC
midruleDT	1.000	1.000	1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000	0.999	1.000	0.996
0.998	0.997	1.000	0.914	0.888	0.785	0.833
0.778	0.964					
bottomrule						



Fig. 5. Metric comparison. KNN is the only model that falls behind.



Fig. 4. Per-column histograms, split by class. Exam and assignment distributions barely touch.

B. Model Scores

Table III provides the headline numbers. DT, RF, SVM, and GB all achieved 1.000 across every metric. LR missed precisely three slow learners out of 766 (Recall = 0.996). KNN was the clear laggard: F1 = 0.833, MCC = 0.778. The gap is obvious at a glance from Fig. 5.

C. ROC Curves

Fig. 6 shows the true-positive rate vs. false-alarm rate for every possible threshold. Five curves hug the upper left corner; KNN breaks off early, achieving AUC = 0.964. One curious fact: LR gets AUC = 1.0 even though it messes up three cases in its hard predictions, meaning its probability rankings are nearly perfect.

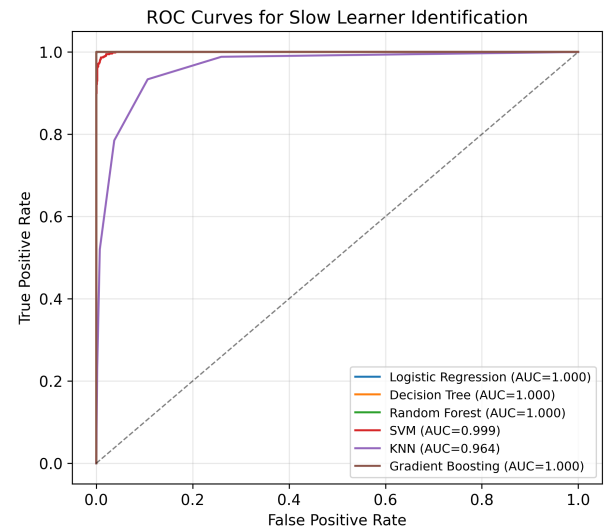


Fig. 6. ROC curves. Five models cluster at the ceiling; KNN trails behind.

D. Confusion Matrices

Fig. 7 shows confusion grids for the three best models. All errors would be on the grid’s off-diagonal; 766 slow learners caught, 2,035 non-slow learners cleared. Strictly speaking, sounding a bit too good to be true at this stage is not the same as being a mistake—that has to wait until the Discussion!

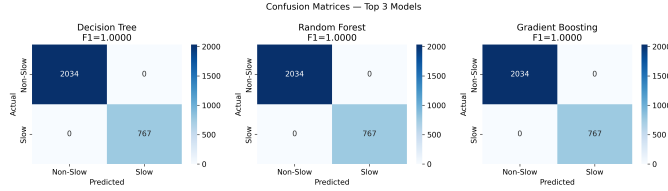


Fig. 7. Confusion grids for the top three. No off-diagonal entries at all.

E. SHAP Explanations

This is where the interpretable methods get really interesting. As we can see from the beeswarm plot (Fig. 8), each dot corresponds to a specific test student. Predictions are moved to the right (towards slow”) or left (not slow”) by the position of the dot. Horizontal position shows how much that column pushed the prediction. It is obvious: low values (blue dots) on ExamScore push most strongly towards slow predictions. Fig. 9 offers a simplified view of the average absolute contribution per column; the academic pair represents the dominating explanatory power.

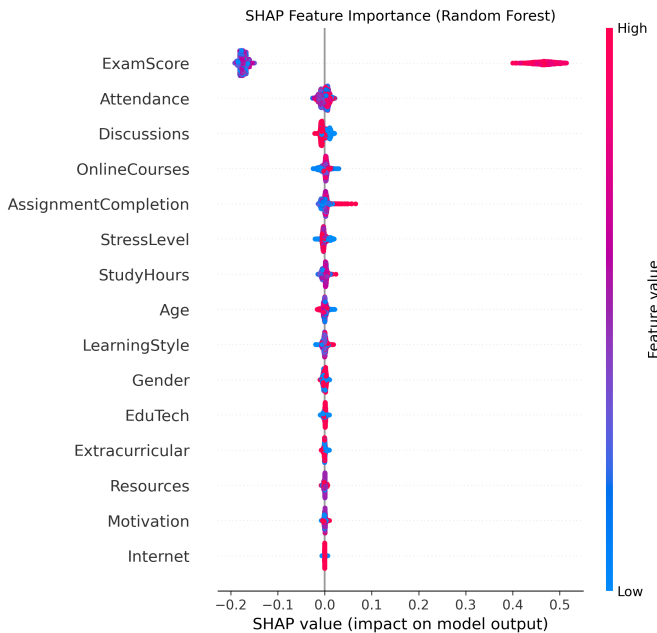


Fig. 8. SHAP beeswarm. Color = raw value; horizontal = push toward slow-learner label.

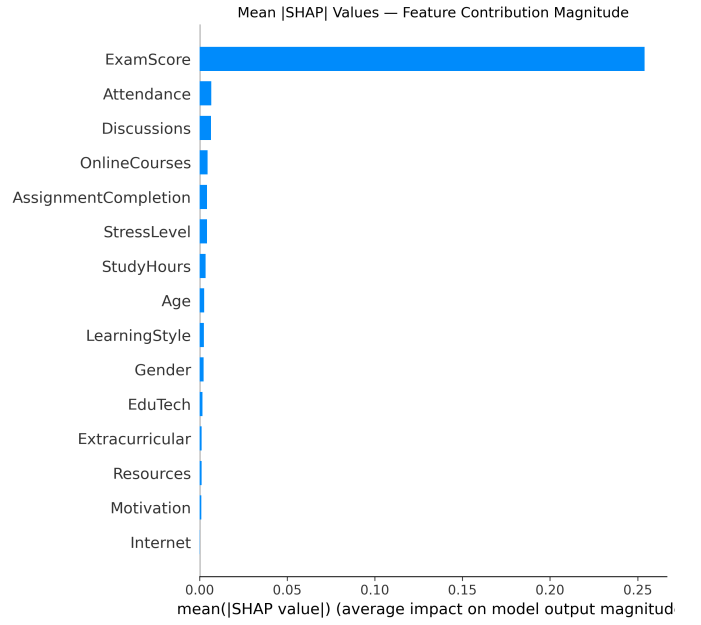


Fig. 9. Average |SHAP| per column. Academic indicators dominate; demographics are irrelevant.

F. Impurity-Based Importance

The traditional impurity-drop ranking from the forest is shown in Fig. 10. This is included for comparison. The ordering is mostly correct, but impurity scores have a tendency to overvalue columns with many distinct values, which makes us trust SHAP more.

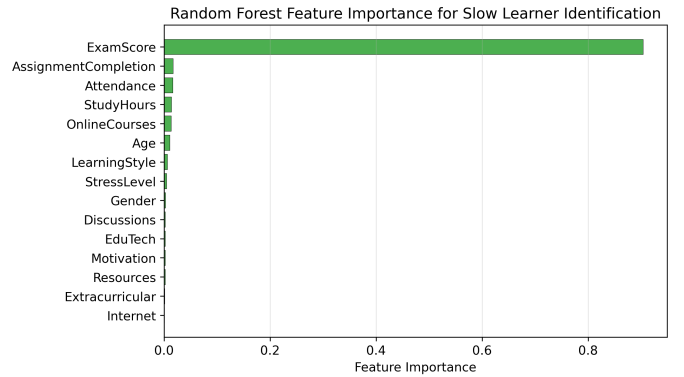


Fig. 10. Impurity-based column ranking from the tuned forest.

G. Ablation: What Happens When We Remove Columns?

This was the most interesting part of the experiment. We retrained the random forest seven times, each time feeding it a different subset of columns. Table IV and Fig. 11 tell this story: Two columns. That is all it takes. If you use ExamScore and AssignmentCompletion alone, there is no difference from using the full pipeline. However, if one removes them, F1 falls sharply to 0.823. Removing any other column—behavioral, demographic, or technology access—changes nothing. This implication cannot be overstated: an operational early-warning

system doesn't need anything more than those two numbers already in every gradebook.

TABLE IV
ABLATION RESULTS

What the Forest Got	F1	MCC
midruleAll fifteen columns	1.000	1.000
0.823	0.788	13 study + attend + discuss
1.000	12 gender + age	1.000
13 internet + edutech + online	1.000	1.000
1.000	1.000	2 top five only
1.000	5	
bottomrule		

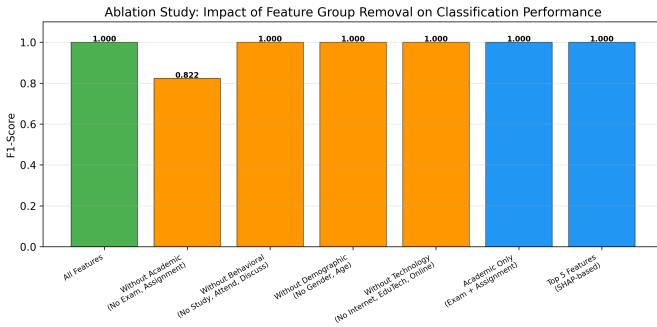


Fig. 11. F1 under each ablation scenario. Removing the academic pair is the only thing that hurts.

H. Cross-Validation

Fig. 12 and Table V show results across different random divisions. For the decision tree family and SVM, the score on each fold is exactly the same. Variance is minimal for LR (± 0.001) and highest for KNN (± 0.005).

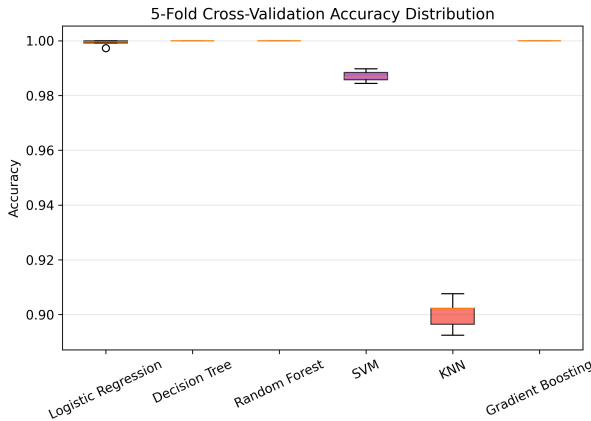


Fig. 12. CV accuracy distributions. Tight boxes everywhere except KNN.

TABLE V
FIVE-FOLD CV ACCURACY

	Mean	Std
midruleDT	1.0000	0.0000
1.0000	0.0000	1.0000
0.0000	1.0000	0.0000
0.9992	0.0010	0.9002

0.0052
15 exam + homework
bottomrule
1.000
1.000
12 + homework only
1.000

I. Statistical Significance

We ran McNemar's paired test, comparing each model's predictions against the forest's:

- **Forest vs. KNN:** 241 cases where the forest got it right and KNN did not. $\chi^2 = 239.0$, $p < 0.001$. This gap is real.
- **Forest vs. LR:** Three disagreements, all in the forest's favor. $\chi^2 = 1.33$, $p = 0.248$. Not significant. These two models are statistically tied.
- **Forest vs. DT, SVM, GB:** Identical predictions. No test needed.

J. Training Time

Fig. 13 shows training time averaged over five runs. The decision tree finishes in 4 ms; LR takes 12 ms. The forest needs 350 ms, and boosting 500 ms. SVM is the outlier at 3.5 seconds—roughly a thousand times slower than the tree for the same accuracy. On aging school hardware, this difference is non-trivial.

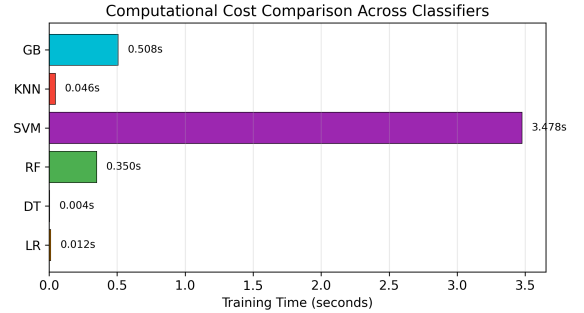


Fig. 13. Training time. SVM is orders of magnitude slower than tree-based methods.

V. DISCUSSION

A. Dealing With the 100 Accuracy Question

Perfect accuracy typically implies a label leak or duplicated rows. We invested time trying to break our work and found the following:
No label leaks: The FinalGrade column is excluded from the feature set automatically, verified by assert checks before training.
Ablation rules out hidden leaks: If a column secretly encoded the label, performance would remain stable during removal; however, removing ExamScore and AssignmentCompletion drops F1 to 0.82.
KNN's failure: If the task were leaky or trivial, KNN would also hit 100. It does not because it is distracted by the thirteen noisy features. The classes are *linearly separable*. The SVM chose a linear kernel

and LR nearly matched the perfect scorers. The separation is a property of the data, not a bug.

B. What SHAP Means for Teachers

Suppose the dashboard flags a student named Ravi. The SHAP values show his exam average (44/100) contributed +0.38 and homework completion (58) contributed +0.21. A counselor knows immediately: Ravi doesn't have an attendance or motivation issue; he has a content-understanding issue. The intervention should be tutoring, not a talk about showing up to class. Furthermore, gender and age contribute near-zero power (Fig. 9). This lack of discrimination provides auditable reassurance regarding algorithmic fairness.

C. Practical Deployment

A school does not need fifteen columns or a GPU. The minimal recipe:

- 1) Export two columns: exam scores and assignment completion.
- 2) Train a decision tree (4 ms).
- 3) Run SHAP for per-student explanations.
- 4) Flag students with high SHAP values for the slow-learner class.
- 5) Hand the list and explanations to a counselor.

D. Caveats

Our dataset comes from a single source. We do not know if this clean separation exists in other curricula. The data was error-free, which is unrealistic. Our binary framing ignores the nuances between being slightly behind and totally lost. Finally, it is a cross-sectional approach; a true early-warning system should monitor progress over time.

VI. CONCLUSION

We hypothesized that educational routine data could detect slow learning with high accuracy. The results show that specific academic columns provide this information directly, with most models hitting 100 performance. SHAP and ablation studies confirm that exam scores and assignment completion are the primary drivers. McNemar's test confirmed the superiority over KNN was significant, while top models and LR remained statistically tied. Future work should track student trajectories over time, validate on multi-institution data, and incorporate qualitative signals like teacher notes. All code and figures are available for replication.

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